Landsat 10 MOC Functions Summary

Version 2.0

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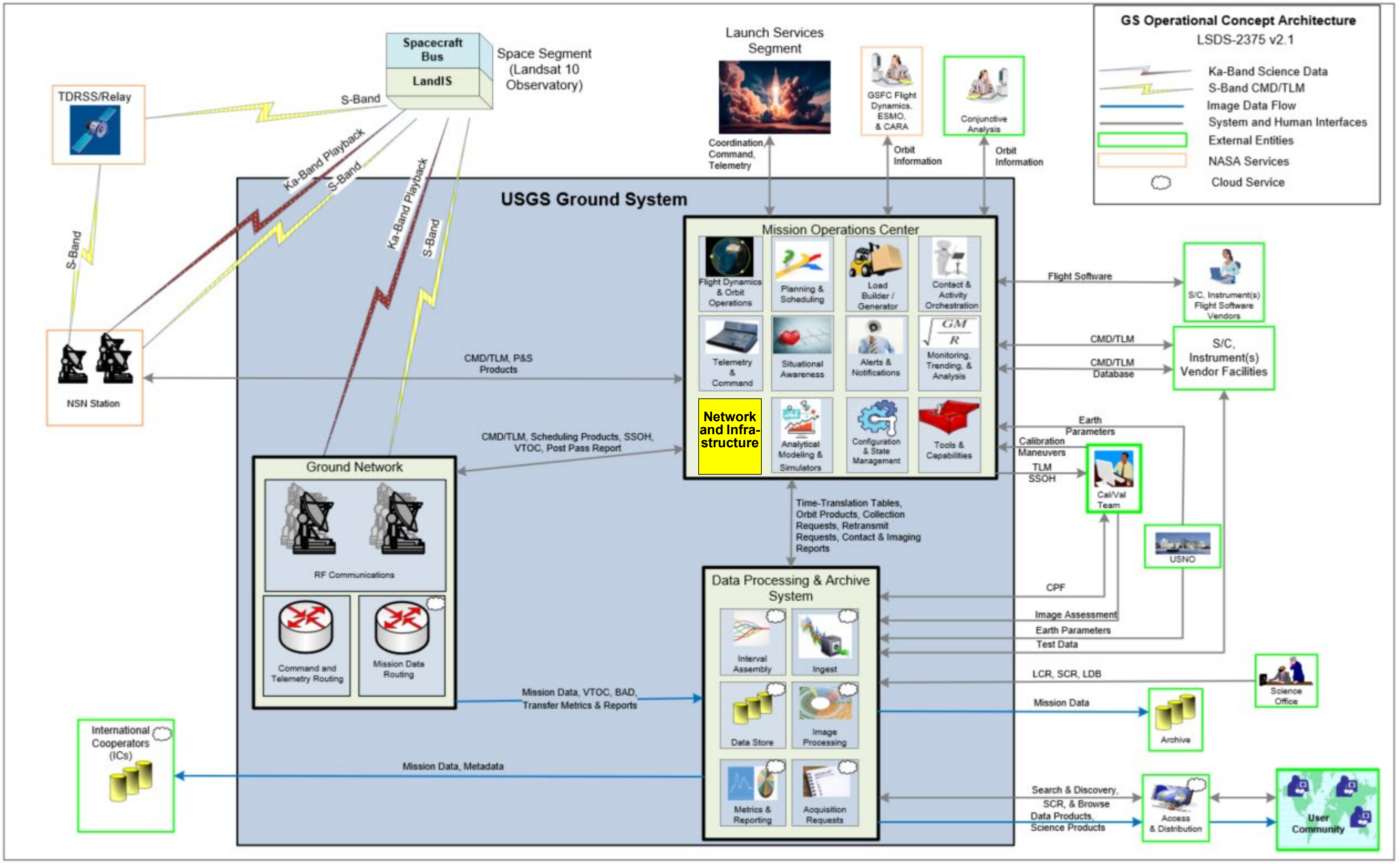
Purpose

- This material is intended to provide a high-level summary of the core functions of the Landsat 10 (L10) USGS MOC and operations and to clarify terminology.
- The functions included here are not intended to be an architecture or even reflect subsystems. USGS recognizes that implementers may refer to these functions using different names and may consolidate or divide functions within subsystems of an architecture in ways not represented here.
- The summary is not intended to be comprehensive but rather to highlight the scope boundaries of the L10 MOC and of each defined function.

L10
Landsat 10

Ground Segment Architecture Summary

Notional Depiction Only



- **Mission Operations Center Functions**
  - **Telemetry and Command:** real time activities, and routine operations
  - **Flight Dynamics/Orbit Operations**: orbit determination, ephemeris and event generation, and maneuver planning (orbit maintenance)
  - **Planning and Scheduling**: contact, imaging, calibration, and all special activities to be executed by the Observatory
  - **Analytical Modeling**: modeling and tools to verify current flight operations concepts and assess potential modifications. **Simulators:** activity validation, anomaly support, training, product updates, etc.
  - **Trending and Analysis**: short-term, and long-term trending
  - **Configuration & State Management**: manage observatory and ground
  - **Contact & Activity Orchestration**: Plan, schedule, and manage all activities in support of lights out operations
- **Ground Network (GN) Functions**
  - Provide the **RF interfaces** between the ground and Observatory
  - **S-Band** interface for command and telemetry, and SSOH
  - **Ka-Band** interface for the capture of mission data from the Observatory
  - Mission **data transfer** to Landsat cloud
- **Data Processing & Archive System Functions**
  - Generation of **special collection requests**, transmission to LMOC
  - **Interval assembly and accounting** of mission data files received from the GN
  - **Ingest** of mission data files to produce Level-0 intervals
  - **Storing and archiving** the data and products (short-term and long-term storage)
  - Level 1 and Level 2 **product generation**
  - User **discovery and access**
  - **International Cooperator** data and products exchange
  - Ground system and science **metric** gathering and reporting

MOC Interfaces

- **MOC/GN**
  - T&C – TLM and CMD data exchange via encrypted WAN connection, i.e. – SSL tunnel
  - Scheduling – Interface to each LGN station to pass contact schedules and ephemeris in a single common format for all stations, includes any handshake protocols to confirm receipt and acceptance of scheduled passes (i.e. – API, CFR/CFC/CCS, etc.)
  - VTOC – S/C mass storage (SSR) directory listing (volume table of contents) of files stored and corresponding file attributes (i.e. – priority, creation date/time, playback status, protected, etc.), nominally via Ka, but download-able via S-Band on demand (i.e. – for anomalies)
  - Post-Pass Report (GPR) – Listing of files that passed and failed checksum verification (e.g., ACK/NAKs), note this may not include leading partial files that are mission file ID information (and thus cannot be identified by the GNE)
  - Status – Station equipment/systems status for use during pre-pass and pass operations to support MOC automated functions (i.e. – for CAO), not expected to allow MOC control of any station equipment/systems
  - Tracking information (i.e. – angle files), for LEOC and contingencies
- **MOC/DPAS**
  - LOSTT, OSTT – Scene to interval mapping table from MOC to DPAS, i.e. – what files are part of a given imaging interval, scheduled acquisitions (from S/C)
  - LCR/SCR – Imaging requests from DPAS to MOC, assume these are initially vetted (by science team) for gross errors and format (also handshake back to DPAS)
  - Retransmit Request – Handshake protocol to receive retransmit requests and verify status of request from S/C
  - Reporting – Provide reporting, metrics and statistics to DPAS for use outside of the MOC secure environment, from MOC to DPAS; also DPAS statistics back to MOC
  - Scheduling Artifacts – Send contact schedules to DPAS for mission planning (i.e. – data volume management, etc.)
  - Deconfliction Reports – May be included as part of other interfaces (OSTT, LOSTT, handshakes, etc. from S/C)
  - Definitive Ephemeris and other FD/OD information from MOC to DPAS as needed (i.e. – fuel usage/monitoring, etc.)
- **Cal/Val**
  - Cal Requests – Two-way coordination to support planning of cal/val maneuvers/acquisitions
  - TLM – MOC TLM access to support analysis of instrument performance and scheduling cal/val activities

MOC Interfaces, cont.

- **MOC/Externals**
  - S/C Vendor EGSE/Facility (“Listen-Line”) – T&C WAN connection to S/C vendor facility to receive copies of all S/C T&C traffic during I&T, also used for MOC to command S/C (and receive TLM) during MRTs and CTs (also supports connections at launch site)
  - Instrument Vendor – Interface to receive simulator updates and pass operational TLM (and CMD) data for troubleshooting and instrument maintenance (products)
  - S/C Vendor – Interface to receive simulator updates and pass operational TLM (and CMD) data for troubleshooting and S/C maintenance (products)
  - Launch Site Interfaces – Connections to S/C via launch vehicle (facility) for TLM (possibly commanding too) during I&T
  - NASA/NSN – NASA NSN/NEN station interface for T&C data transfers, scheduling and status, could be multiple interfaces or single NexTEra-type unified interface
  - NASA/SN – NASA SN/TDRSS interface for T&C data transfers, scheduling and status, could be multiple interfaces or single NexTEra-type unified interface, availability of NASA SN resources TBD for long-term ops
  - NASA Commercial SN – T&C data transfers, scheduling and status interface with commercial SN (in-space relay) provider for long-term ops, could be multiple vendors, interface could be unified NASA interface or vendor-specific
  - NASA FDF – Tracking and OD information during commissioning, also for contingencies as needed
  - Conjunction Assessment – Two-way interface with conjunction assessment provider, expected to be NASA CARA during commissioning and transition to a commercial provider for ops, DoC/TraCSS
  - Status – High-level MOC (and limited S/C) status, e.g. – web GUI displaying MOC status for approved users (but could also be public-facing)
  - US Space Command (Space Force) – 18th/19th space defense squadron to coordinate CA information; send TLM to UDL
  - US Naval Observatory – Timing information (e.g. – leap seconds, etc.)

MOC Functions and Capabilities

Landsat 10 MOC Functions and Capabilities

Mission Operations Center (MOC) Functions/Capabilities

Telemetry & Command (T&C)	Flight Dynamics/ Orbit Operations (FD/OO)	Planning & Scheduling (P&S)	Contact & Activity Orchestration (CAO)
Configuration Management & State Management	Trending & Analysis	Alerts & Notification	Load Builder/ Generator & Validation
Analytical Modeling & Simulators	Network and Infrastructure	Mission Operations Situational Awareness	Other Tools/ Capabilities

External Interfaces

- Ground Network (GN)
- Data Processing & Archive (DPAS)
- Cal/Val
- S/C Vendor
- Instrument Vendor
- Launch Services Segment
- NASA GSFC Services:
  - Flight Dynamics Facility (FDF)
  - Conjunction Analysis (CARA)
  - Near Space Network (NSN)
- US Space Command
- Status Dashboard
- Conjunction Analysis Services
- US Naval Observatory
- DoC/TraCSS

Telemetry and Command (T&C)

- **Core/Basic/Common Capabilities**
  - Telemetry processing/decommutation/engineering conversion/distributing to MOC subsystems/etc. and Command processing/assembly/transmission/confirmation/etc.
  - All standard functions needed to autonomously manage real-time station contacts (pre-pass, pass, post-pass activities), including managing connections with stations and automated failovers as needed
  - Automated and user directed contact activities, including procedure execution, load uplink, telemetry limit monitoring, user page/plot display, archive of all activities performed, etc.
- **Landsat 10 specific required capabilities and highlights:**
  - Command and Telemetry Database will be CCSDS XTCE Standard
  - Ability to split/manage large loads that span multiple contacts
  - Execute mission data retransmit requests provided by DPAS
  - Perform Observatory state and configuration checking/monitoring (using received TLM)
  - Command encryption, to potentially include Software command encryption (in the cloud)

  - Note: Landsat separates Tracking from Telemetry and Commanding (or Command & Control). Landsat used GPS for Orbit Determination, and NASA FDS for Tracking as a backup only.
- **Desired capabilities (may not be required):**
  - Apply user generated T&C database updates and/or limits (allows for limit/alarm behavior as a function of ops mode, e.g. maneuvers)

Flight Dynamics / Orbit Operations (FD/OO)

- **Core/Basic/Common Capabilities:**
  - Orbit Determination: Use S/C GNSS telemetry to create definitive orbit; Ingest external tracking data as needed (i.e. NASA FDF, commercial tracking services, angle tracking data from GN stations, etc.) for backup orbit determination capability
  - Orbit Propagation: Generate and provide ephemeris for all ground and operational needs
  - Orbit and Maneuver Planning
    - Assess adherence to orbit and mission parameters (ground track, inclination, constellation spacing, drag monitoring, fuel management, etc.)
    - Generate and assess (validate) ephemeris and maneuver plans as needed for orbit maintenance, mission safety (i.e. RMMs), and one-time and special events in cooperation with S/C vendor and mission engineering (i.e. LEO, constellation respacing, EOL deorbit, etc.)
- **Landsat 10 specific required capabilities and highlights:**
  - Generate, Maintain, and Deliver definitive ephemeris (historical), and other relevant mission parameter to DPAS as needed (TBD)
  - Event Generation: Generate orbital events for GS use (contact in views for scheduling, sun/eclipse, moon angles and times for calibration, descending node crossing, solar calibration angles, etc.)
  - Flight Dynamics
    - Perform attitude and ACS performance monitoring and management (if not done by T&A) - (e.g. MTASS on L8/L9)
    - Attitude Slew planning (e.g. ACS or Science Calibration, maneuver slews, etc.) and keep out zone monitoring (Sun in sensor FOV, Sun/Earth in Radiator FOV, etc.)
  - Interface with external conjunction assessment (CA) services (i.e. NASA CARA, SpaceNav, etc.) for Space Situational Awareness and coordination of risk mitigation maneuver (RMM) planning (and execution)
- **Desired capabilities (may not be required):**
  - More dynamic drag model – above standard drag model – things like solar weather induced changes, dynamic models, etc.

Planning and Scheduling

- **Core/Basic/Common Capabilities:**
  - Ground network (GN) and In-Space Relay (e.g. TDRSS or similar) contact planning, scheduling, deconfliction, etc.
  - Routine Observatory activity planning, deconfliction, constraint checking, and validation
    - Includes contact activities, Cal/Val activities, attitude and propulsive maneuvers, engineering requests, routine maintenance, etc.
  - Creates Observatory activity schedule for command load builder/generator
- **Landsat 10 specific required capabilities and highlights:**
  - Maintain the ground imaging plan and schedule (defined in the LCR, and modified by any active SCRs)
  - Predict Observatory mass storage device usage and schedule necessary ground contacts to downlink expected science data, including adding new contacts if needed (due to missed downlink, or other issue)
    - Note: this could be a function of the Analytical Modeling capability that is used as input for contact and schedule updates.
  - Ingest, validate, and deconflict SCRs and LCRs received from DPAS
  - Validate the Observatory provided imaging plan (the 18-day onboard WRS-3 scene (Path/Row) plan) against the LCR/SCRs
  - Autonomously review and validate the Observatory provided 48 hour integrated onboard timeline (IOT) and notify engineering of any issues, errors, or unexpected results
  - Optimization of GN contact schedules while still meeting contact time requirements
- **Desired capabilities (may not be required):**

Contact and Activity Orchestration (CAO)

- **Description / L10 specific capabilities and functions:**
  - Plan, schedule, and manage all routine contact activities (i.e. What activities and commanding will occur on which contact)
  - Accept and validate automated retransmit requests from DPAS, convert to CMD sequence for U/L to S/C, schedule execution of commanding
  - Facilitate review, approval, and execution of engineering approved activities (e.g. command authorization meeting approves commanding for upcoming activity, activity sequence generated, reviewed by engineering team, approved, and scheduled for execution on upcoming contact)
  - Monitor status (contents) of onboard mass-storage system, and estimate mass-storage usage (predict usage based on scheduled imaging, update to “known truth” as VTOCs are received) ← could be part of analytical modeling

  - Note: This can be done in other MOC systems or mission enabling tools or similar. Planning/Scheduling, Load Builder/Generator, T&C, a Mission Management system, JIRA, Pass Plan Tools, Timeline/Activity Tools, etc.
  - Note: On L9, this was/is done in custom tools, Mercury (load review, approval, distribution), JIRA/Mission Operations Timeline Execution (MOTE) (pass plan activity management, procedure review, etc.), and the LEOC timeline tool (no longer used)
- **Desired Capabilities (may not be required):**

Configuration Management & State Management

- **Configuration Management**
  - Landsat uses this in the traditional sense – managing all configured items for the ground system, operations, and Observatory, and has formal processes and tools for updating and changing them. This includes utilizing common tools like JIRA, Confluence, Sharepoint, Clearquest, or others.
  - Includes hardware lists/configurations, ground software (MOC, tools, etc.), Mission Operations Configured Items (Command Procedures, SOPs/COPs, Reference Docs, Loads, FSW images, Onboard parameters, etc.)
- **State Management**
  - This has sometimes been lumped into CM, but for clarity, this is intended to differentiate the ability to track, manage, and compare the specific state/configuration of the Observatory and related ground system components.
  - Observatory & Ground System State Management
    - Monitor, manage, and archive states and configurations for flight H/W, flight S/W, and MOC systems
    - Ability for tracking configuration of the Observatory and ground system and be able to look back to any point in time to prior configurations.
  - Memory management, tracking, and comparison of ground “gold copy” (e.g. disk image, EEPROM, RAM, etc.) of onboard FSW, tables, and parameters
  - Note: Other missions have used "Config Check", "Config Monitor", "State Check" to describe this capability.
  - Examples include:
    - Tracking and checking any onboard parameters (e.g. FSW tables/parameters, load versions, etc.) or states (e.g. Side A "on", onboard process "enabled", etc.) and then logging and taking action based on changes discovered.
    - Checking/auditing the parameters onboard match the ground system models (e.g. Earth parameter definitions for onboard ACS and ground Flight Dynamics models, etc.).
    - Confirming pre-anomaly and post-anomaly Observatory configurations match (i.e. to confirm full anomaly recovery).
- **Desired Capabilities (may not be required):**
  - Digital twin

Trending & Analysis

- **Core/Basic/Common Capabilities:**
  - Process and store Observatory and MOC housekeeping telemetry (health and status) and provide a full suite of trending and analysis tools on all required timescales (near-real time, periodic, long term, and life of mission)
  - Automated statistics and plot/table/report generation for traditional daily, weekly, monthly, quarterly, yearly, and LOM, and user initiated on-demand and custom generation
- **Landsat 10 specific required capabilities and highlights:**
  - Managing, tracking, and reporting key performance parameters of the spacecraft, instruments, and ground system for the life of the mission, including trends and margins
  - Managing key life limiting budgets (e.g. power, fuel, rotational components, etc.), including predictions of failure
  - Life of mission performance tracking
  - Remote access for engineering & Cal/Val teams to support trending & analysis and anomaly resolution
  - Store data in a format and data structure to facilitate AI/ML and other automated analysis tools (i.e. “analysis-ready data”)
  - Execute automated analysis tools (i.e. AI/ML) to monitor large amounts of telemetry points of L10 constellation for anomaly identification at a minimum
    - AI/ML capability is growing and changing rapidly, so requirements shouldn't be tied to specific current implementations, but be as future proof as possible (i.e. “analysis-ready data” vs. specific ML algorithms)
- **Desired capabilities (may not be required):**
  - Provide read-only access to TLM Database for USGS ground system elements (i.e. cal/val, science, DPAS) for data analysis. Clearly identify what data is export controlled

Alerts and Notifications

- **Description / L10 specific capabilities and functions:**
  - Monitors Observatory and MOC performance and status and sends alerts/notifications
  - Delivers messages from MOC applications/subsystems/functions to external systems and operations staff, utilizing configurable settings
  - External alerts to staff via E-mail, SMS text, voice calls, etc.
  - Messages could be machine-to-machine communication (e.g. post outage status to website, send partner system at LGN a message via API, etc.) or
  - Populate, display and update (manage) mission “dashboard” status display ← Could be part of Operations Situational Awareness

- **Desired Capabilities (may not be required):**
  - Message/alert remote acknowledgement by recipient (closed loop)
  - Alert persistence adjustable/configurable by operations staff
  - Track notification times and response times for metric gathering and escalation (e.g. notification not responded to within 10 minutes, so send to next level recipient)

Load Builder/Generator and Validation

- **Description / L10 specific capabilities and functions:**
  - Takes activity schedule created by Planning & Scheduling and converts activity sequences into CMD loads (ASCII to binary) for U/L to the S/C (formatting for U/L and use by the S/C)
  - Load checking and validation (i.e. an independent command load review/validation capability, with user-configurable modules for revised operations and new signatures discovered during mission life).
  - Segments loads into smaller chunks appropriate for command upload windows and creates ability to load large loads over multiple contacts

  - Note: These functions could be separate tools or part of another function (i.e. T&C, CAO, or planning & scheduling).

- **Desired Capabilities (may not be required):**

Analytical Modeling and Simulators

- **Description / L10 specific capabilities and functions:**
  - Analytical Modeling
    - Provides modeling and tools to verify current flight operations concepts and assess potential modifications to
    - Model "what-if" scenarios for different parameters in image scheduling, GN station distribution, etc.
    - Includes, but not limited to:
      - assessing impacts of GN changes to on-board data storage management
      - Imaging acquisition changes
        - e.g. evaluate a request for a new night collection, or a change to the LCR
      - Develop and maintain a model of S/C mass storage operation and file system usage ← could be part of CAO
  - Simulators
    - Provide MOC system interface to simulators provided by S/C and instrument vendors (i.e. T&C and status/control interfaces, etc.) used for testing, training, CMD planning, anomaly investigations, or other engineering activities (i.e. what-ifs, etc.)
    - Note: The Flight Operations Team will operate and configure simulators provided by S/C and instrument vendors , and will provide maintenance (in cooperation with S/C and instrument vendors)
  - Note: These functions could be separate tools or part of another function (i.e. T&C, CAO, or mission planning).
- **Desired Capabilities (may not be required):**

Network and Infrastructure

- **Network and Infrastructure**
  - Establish and maintain equipment, if any, necessary:
    - To establish MOC environments for Operations, I&T, etc
    - For MOC environment interfaces and external interfaces
    - For access to Cloud based functions and services
  - Note: Scope includes all MOC-owned equipment up to demarcation of telecommunications service provider
  - Manage infrastructure and environment (e.g. cloud, infrastructure as code (IaC), containerization, etc.)
- **Software Deployment**
  - Rapid Deployment
  - Automated testing in I&T – standard automated regression testing
- **MOC IT Environment**
  - Support and maintain tools, etc
  - Supports MOC Data Management for optimized data storage in a way that enables future data analysis tools (see Trending & Analysis)
- **Security Scanning & Patching Management and SCRM/Threat Management**
  - Support IT security and patching
  - Ground security council – establish security agreement for patching ground hardware/software
  - Ability to easily revert back
  - Periodic checks against known bad actor lists throughout life to confirm existing HW and SW hasn't been compromised (sold to a banned vendor/country or become a known unsecure vendor, etc.)

Mission Operations Situational Awareness

- **Description / L10 specific capabilities and functions:**
  - User-configurable dashboard for a centralized display, providing overall status of the Observatory, MOC, and USGS Ground System for health monitoring, metrics, highlights
  - Drill down capability from dashboard; can easily dig down into the subsystems underneath status to investigate and troubleshoot

- **Desired Capabilities (may not be required):**
  - Customizable dashboard with ability to integrate external tools. Examples include tools for near-term forecasts for space weather (from NOAA or Space Force), Space Weather anomaly assessment tools, CA reports/dashboard, drag models, RF interference (from Space Force)

Other Capabilities

- **Incident Management Tool & Process**
  - Internal MOC system development incident management tool – MOC dev vendor
    - Note: Project will select the incident tool for GRTs, MRTs, ops product development, etc.
  - Operational incident management - work with the Government to define tool and process.
- **Collaboration Tools**
  - Can handle multi-agency CUI, ITAR data, meets security guidelines